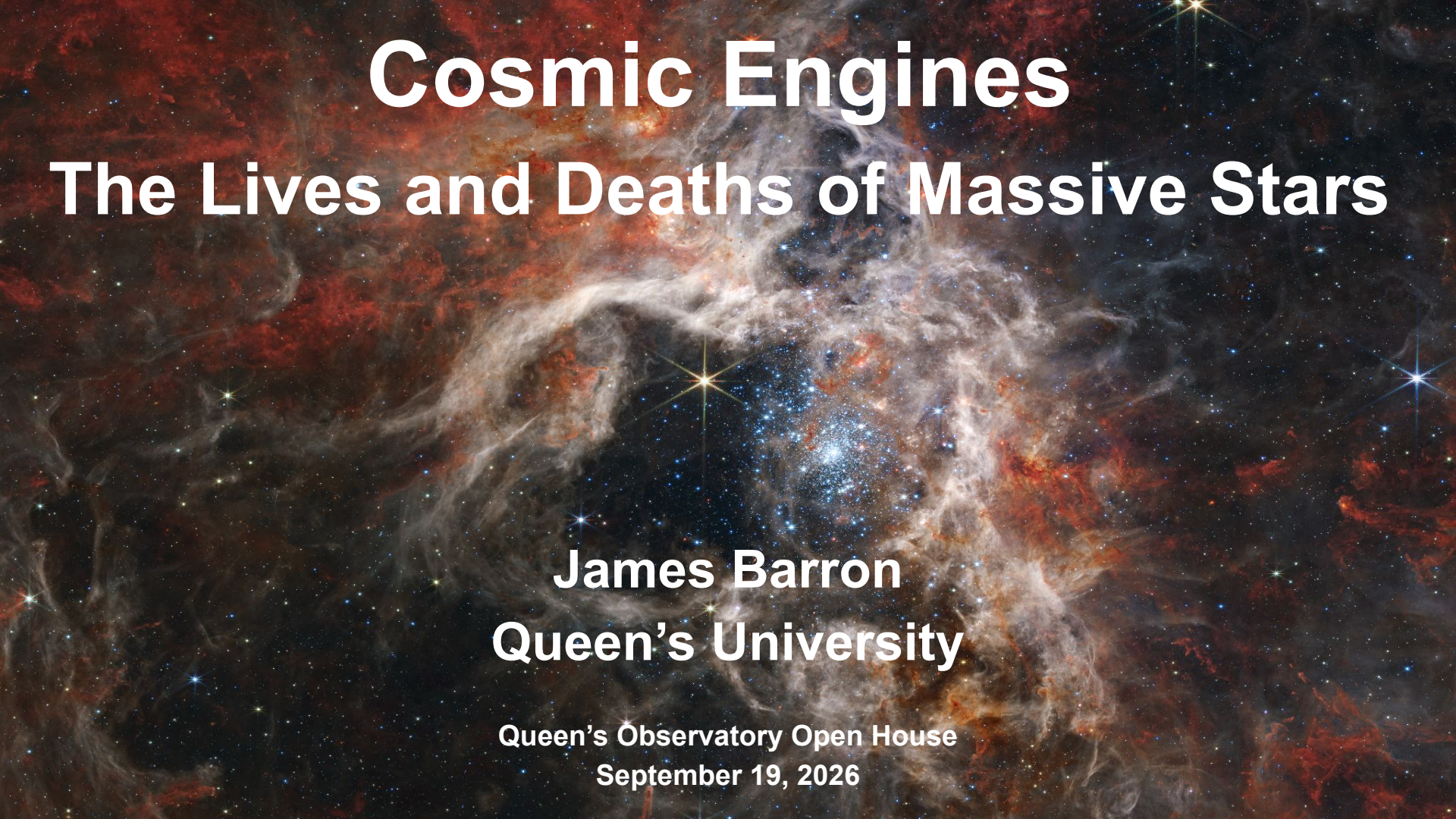




Cosmic Engines

The Lives and Deaths of Massive Stars

James Barron
Queen's University

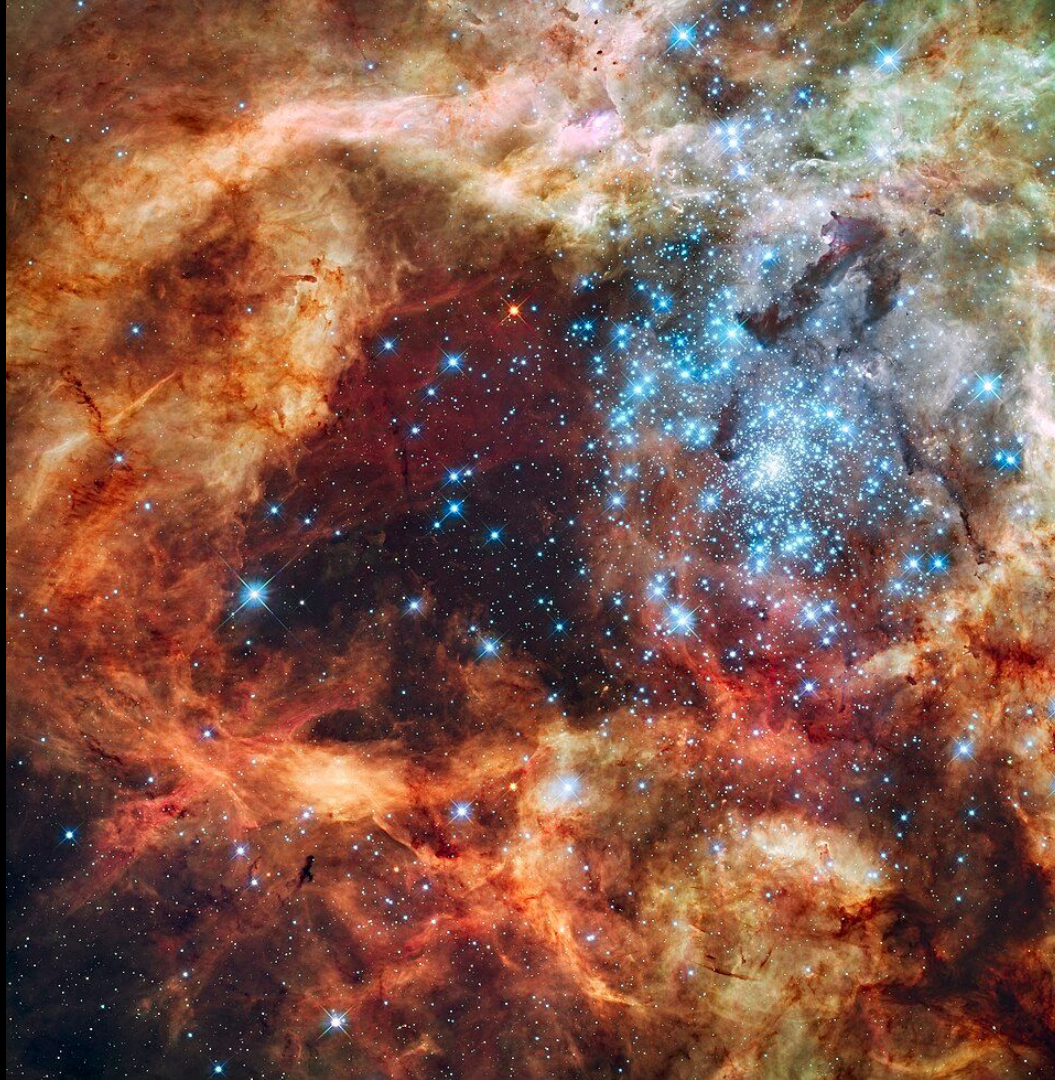
Queen's Observatory Open House
September 19, 2026



Tarantula Nebula

Image: NASA, ESA, CSA, STScI, Webb ERO Production Team

R136



Who am I?

Who am I?

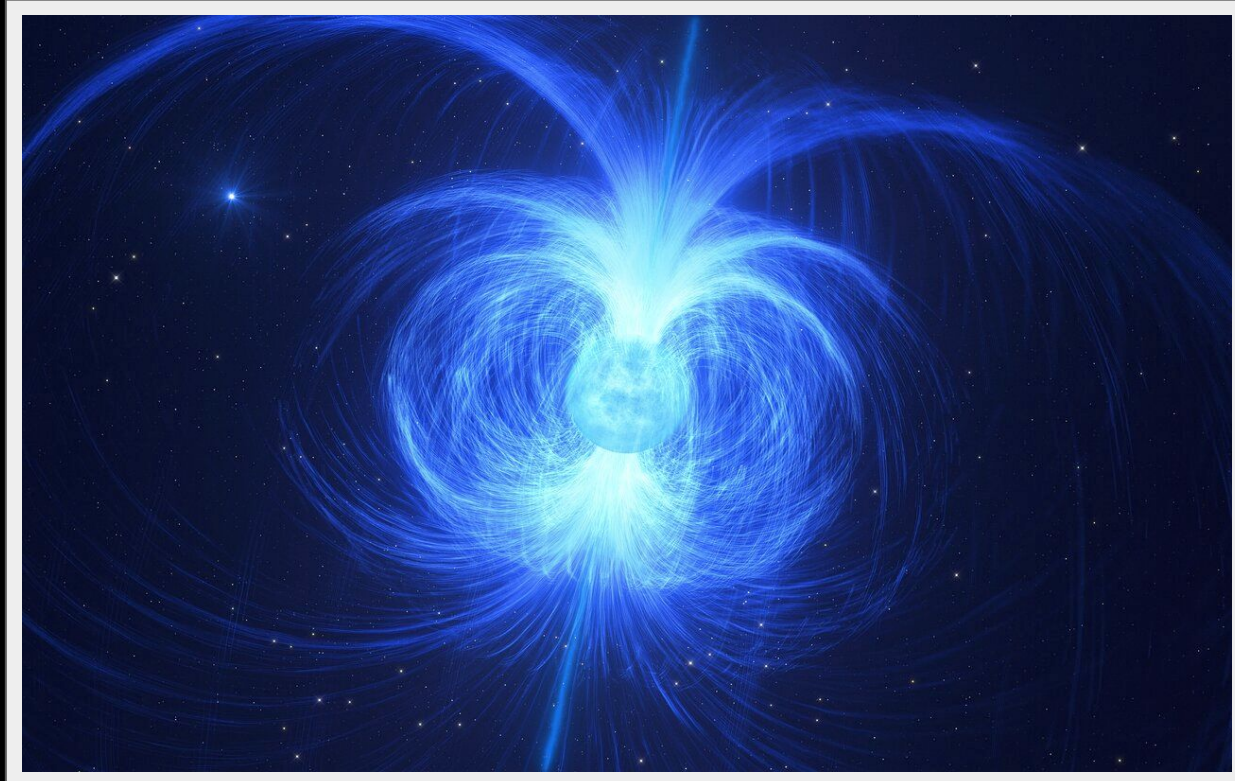
Observational Stellar Astrophysicist



What do I study?

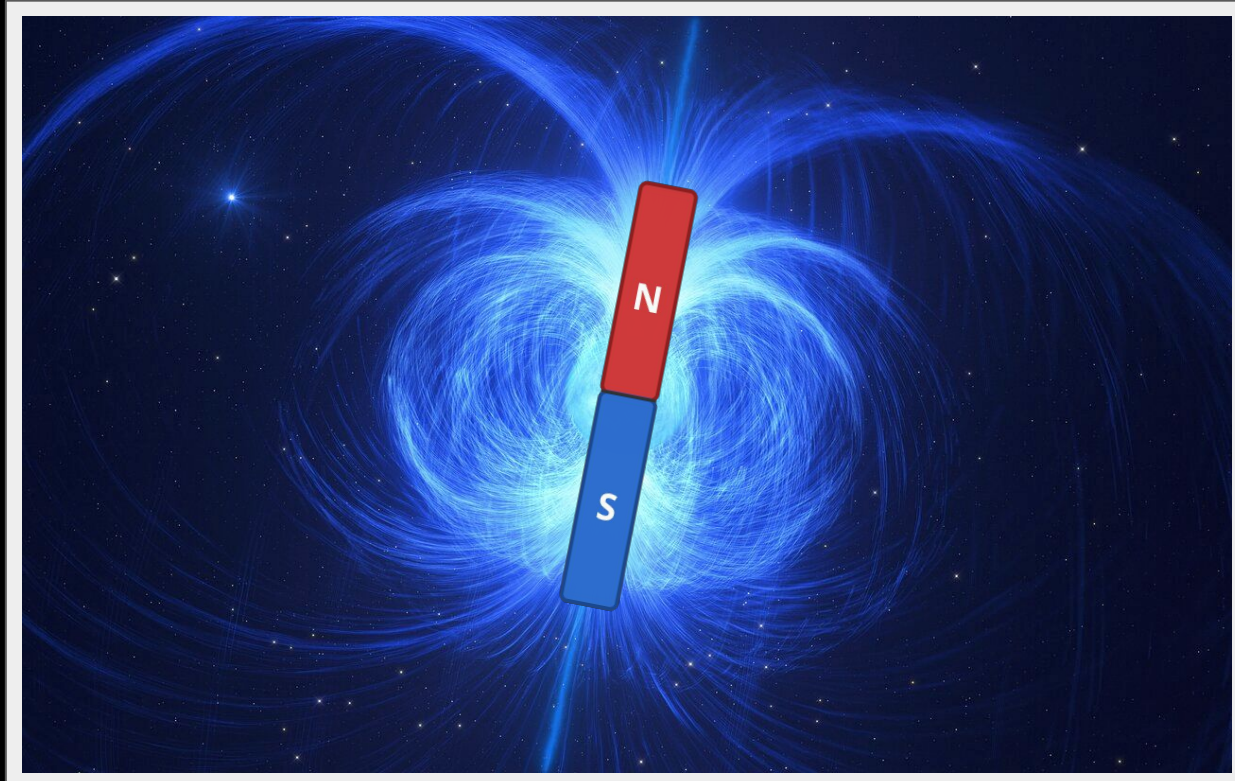
What do I study?

Magnetic Massive Stars!

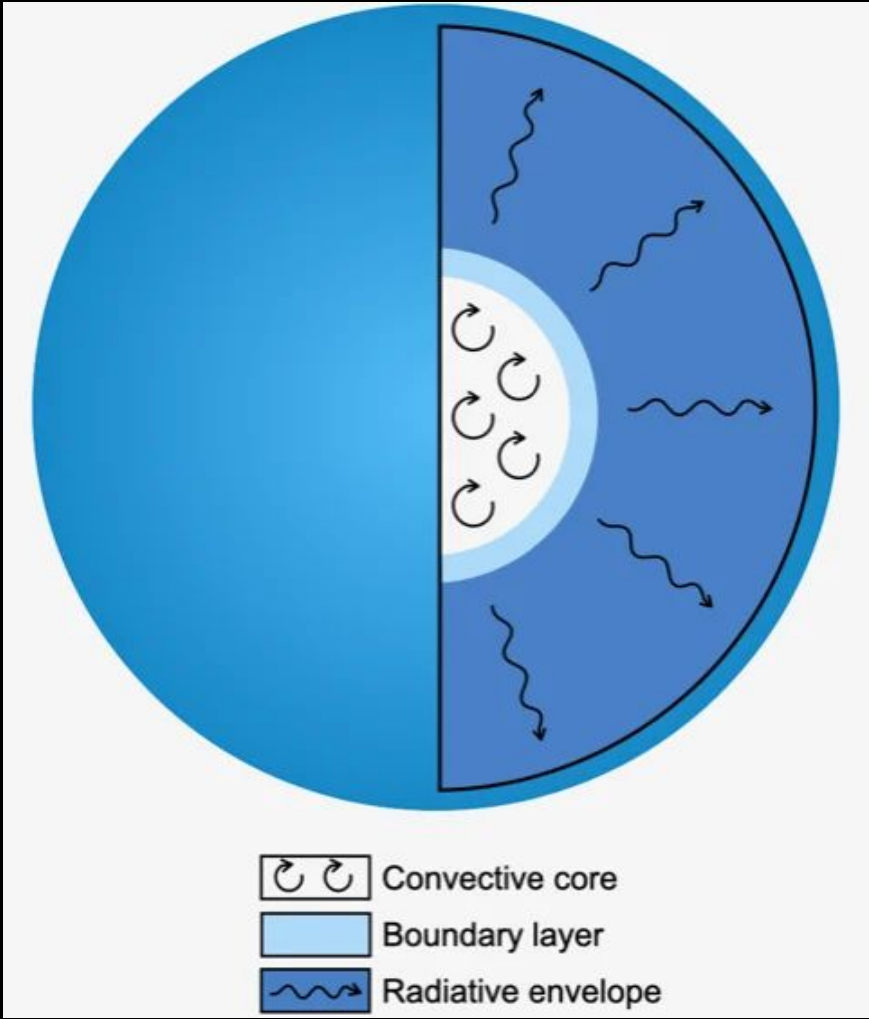
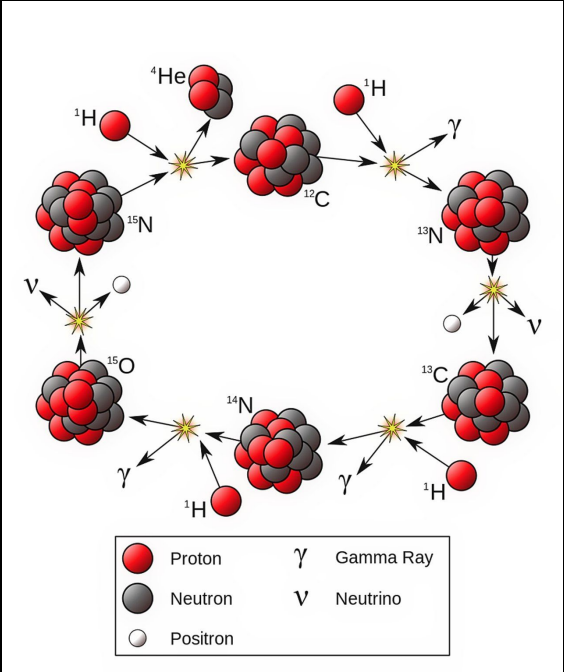


What do I study?

Magnetic Massive Stars!



**What is a
massive star?**

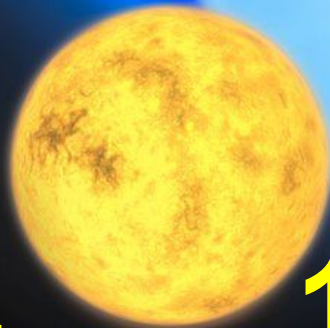


Massive
stars are
heavy.



200x

8x to 60x



1

0.5x

| red dwarf

| yellow dwarf (Sun-like)

| blue dwarf

| R136a1

Massive
stars are
born **hot**.



15,000 - 45,000°



6000°



3000°

| red dwarf

| yellow dwarf (Sun-like)

| blue dwarf

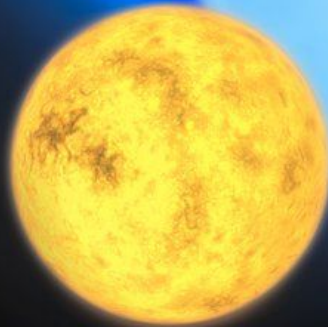
| R136a1

Massive
stars are
'young'.



1 – 4 Million

4.6 Billion



| yellow dwarf (Sun-like)

| blue dwarf

| R136a1



| red dwarf

1 – 10 Billion

Massive
stars are
rare.



< 0.1%

10%

R136a1

blue dwarf

yellow dwarf (Sun-like)

red dwarf

75%



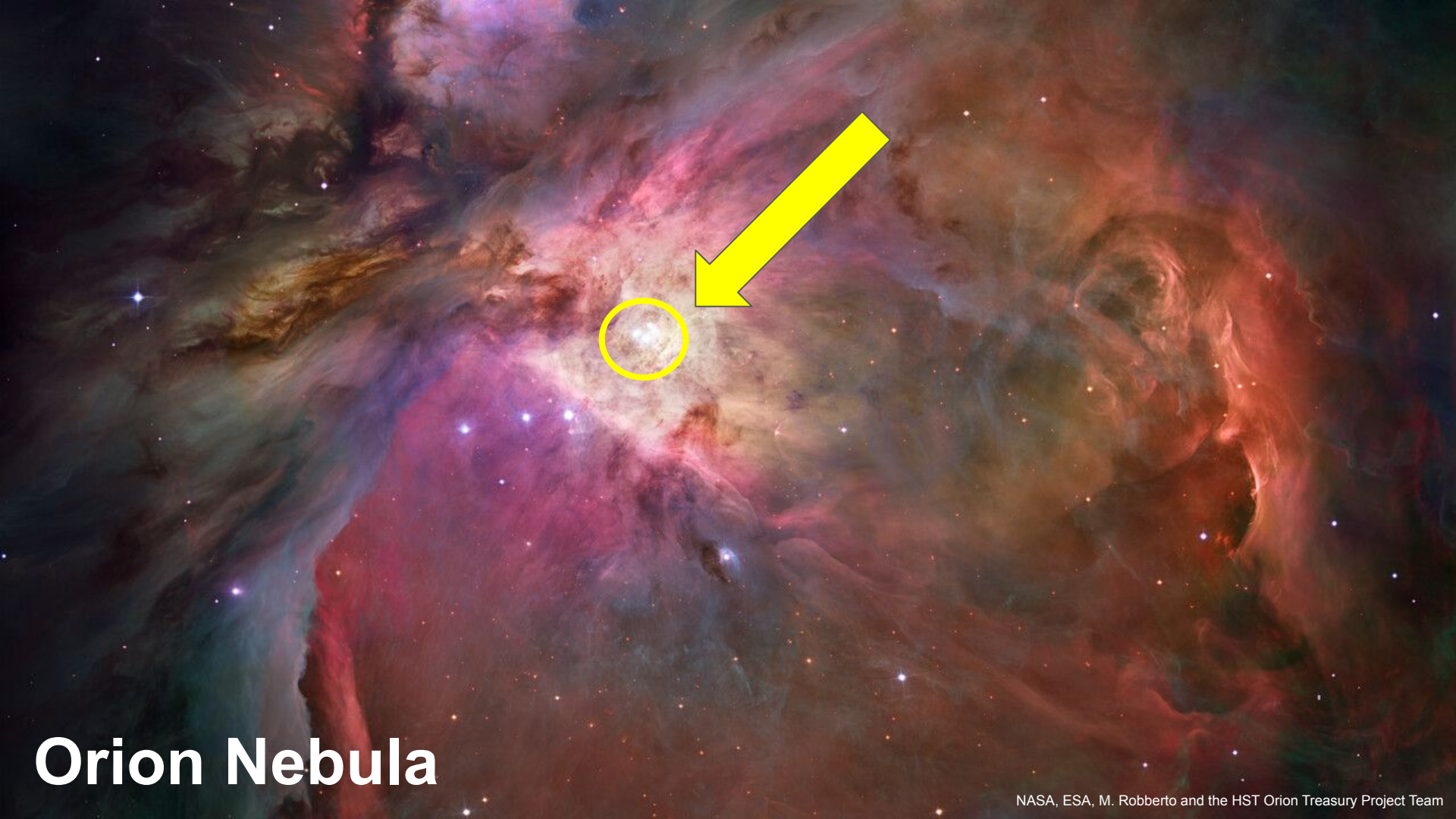
Whirlpool Galaxy

NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)



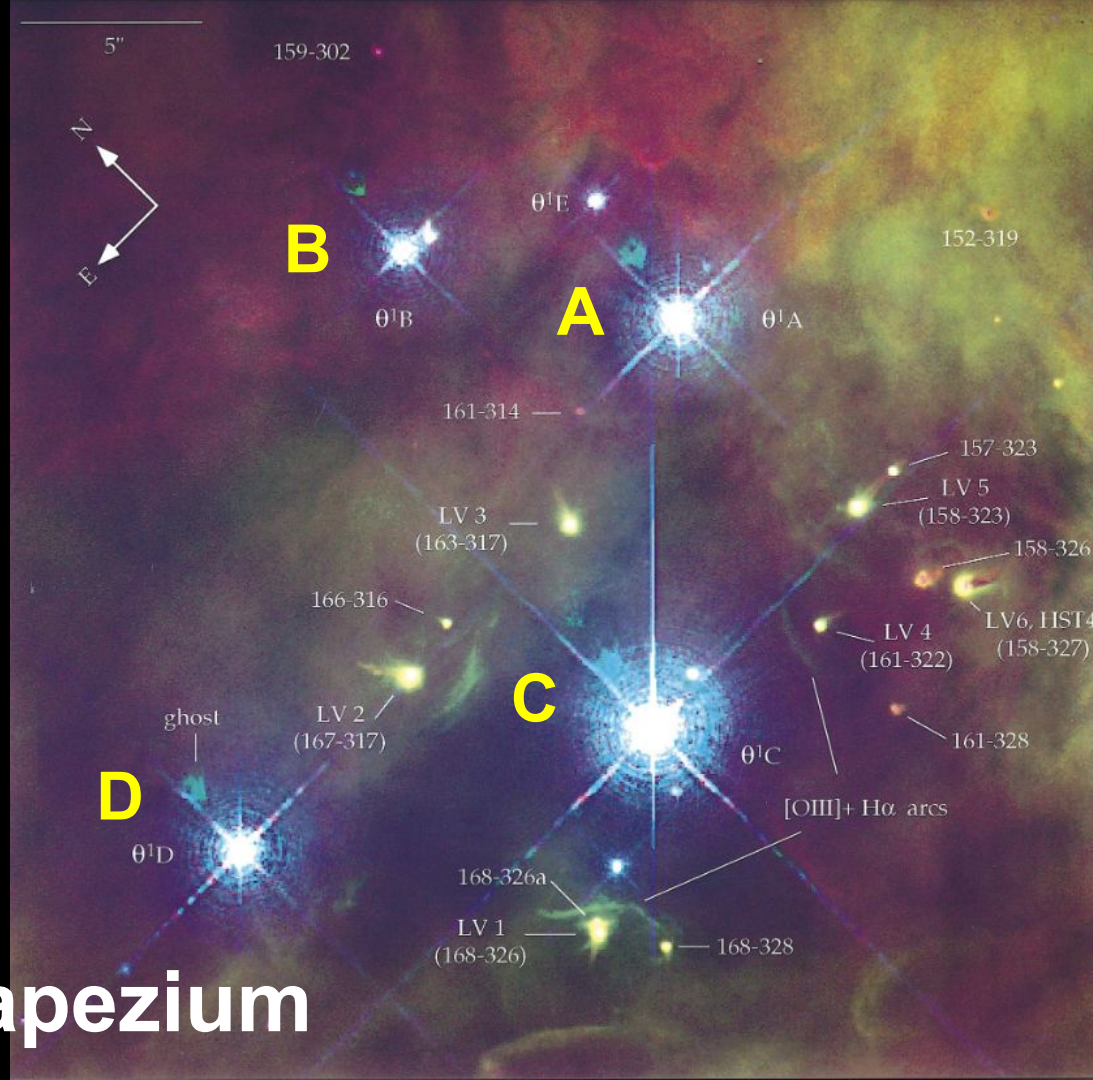
Orion Nebula

NASA, ESA, M. Robberto and the HST Orion Treasury Project Team

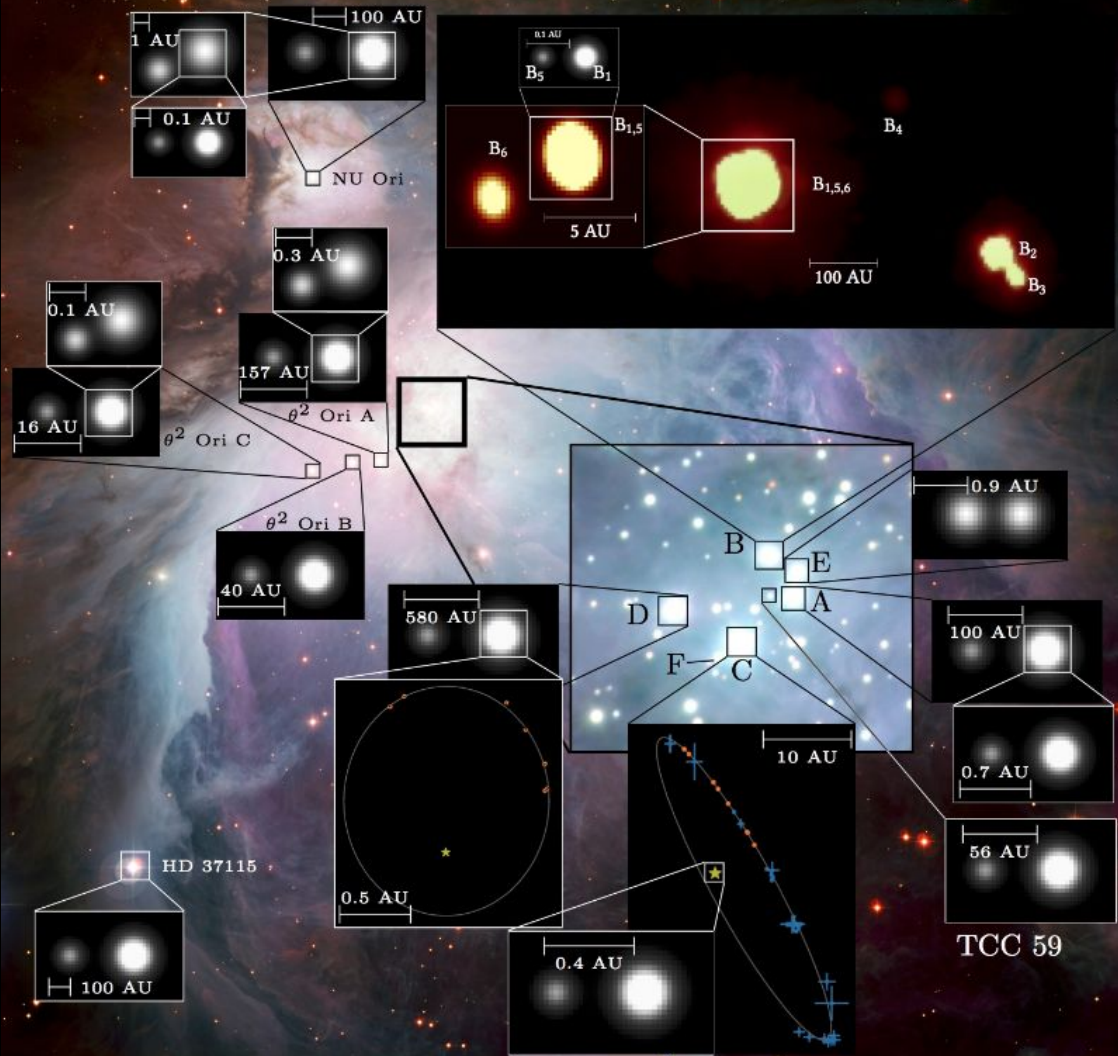


Orion Nebula

Orion Trapezium

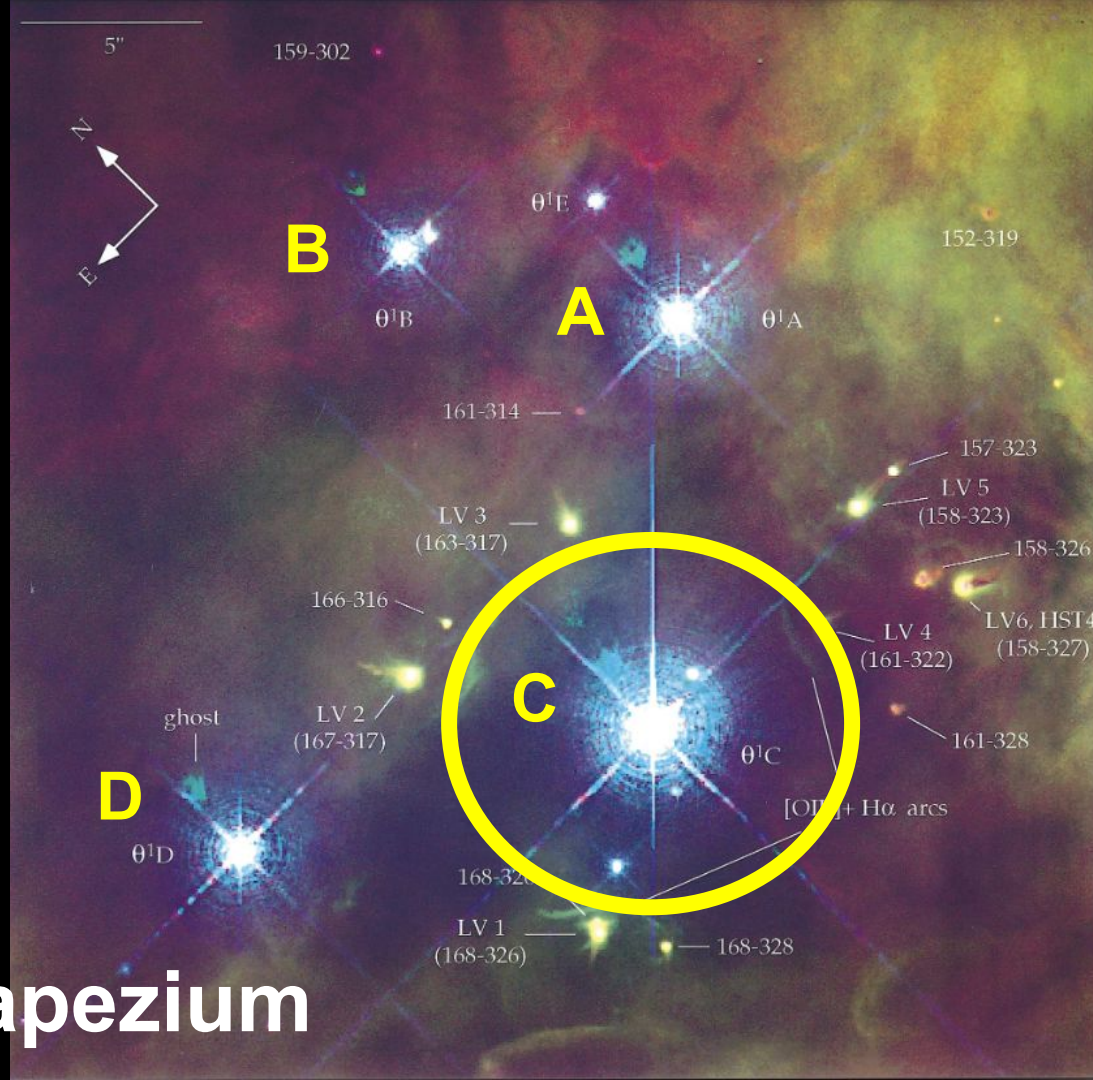


**Massive stars are
born as **twins!**
(or triplets or
quadruplets or
quintuplets...)**



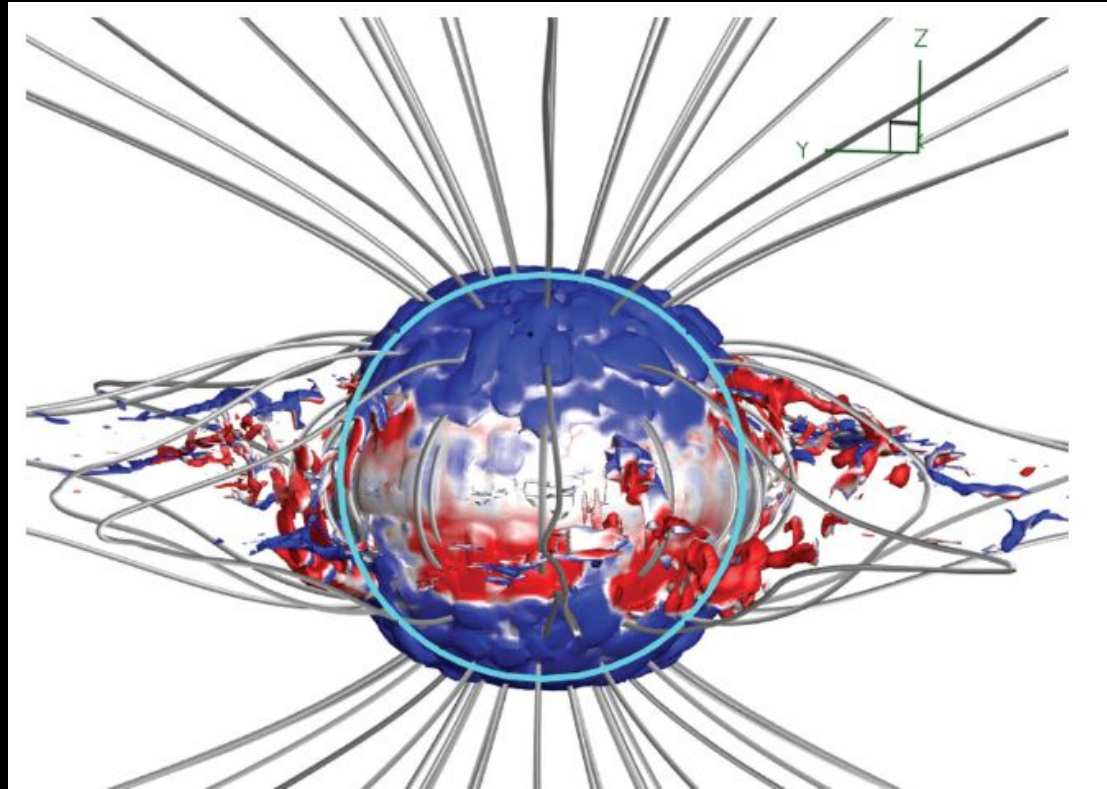
Star Formation Simulation





Orion Trapezium

- Confirmed magnetic in 2002.
- 1000x stronger magnetic field than the sun.
- Only 12 known similar stars.

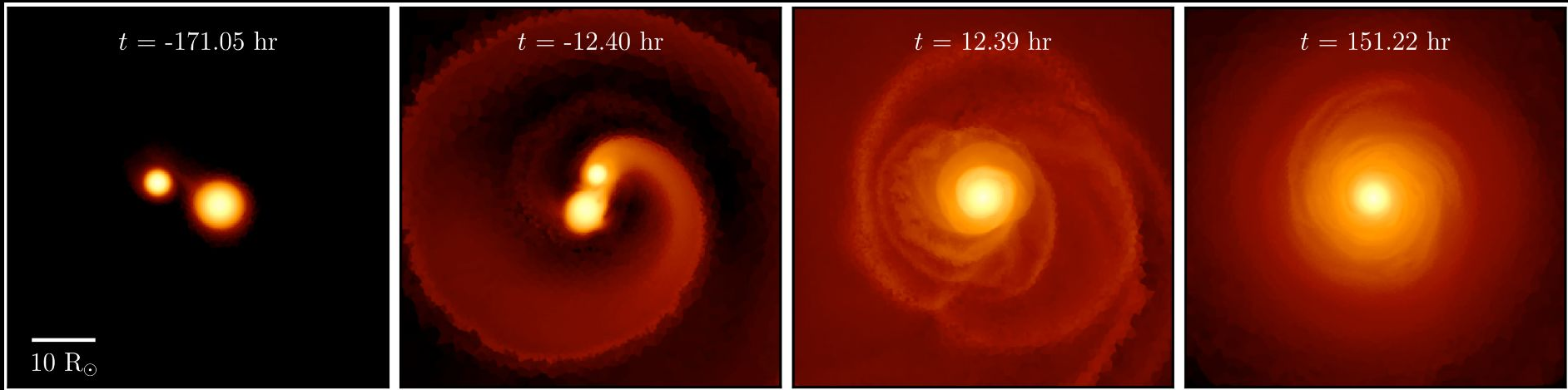


Θ^1 Ori C

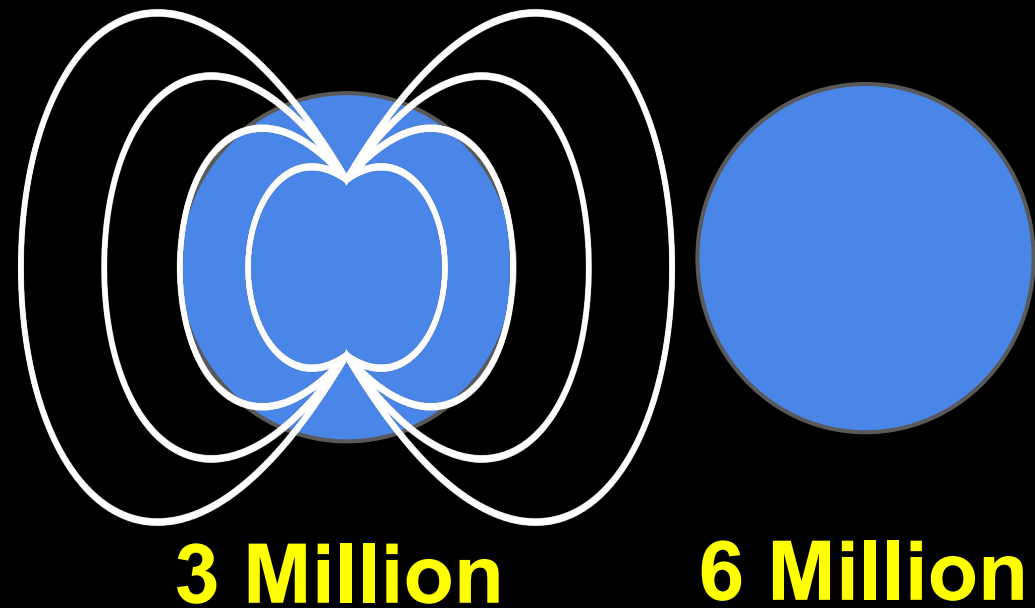
How do you make Θ^1 Ori C?

How do you make Θ^1 Ori C?

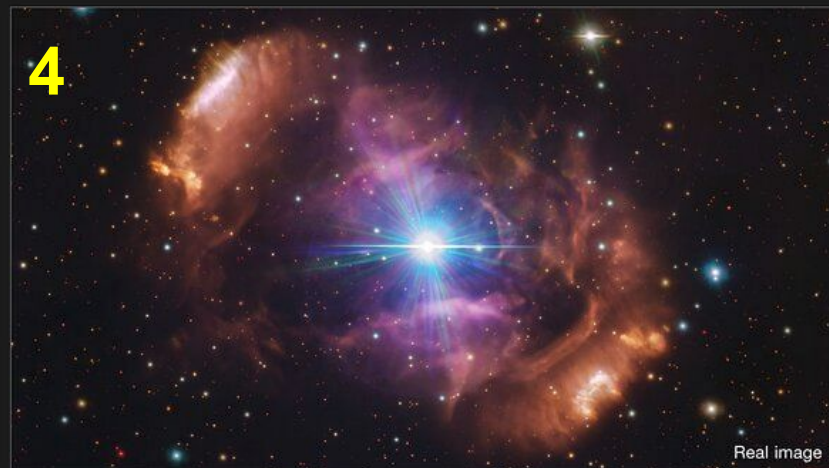
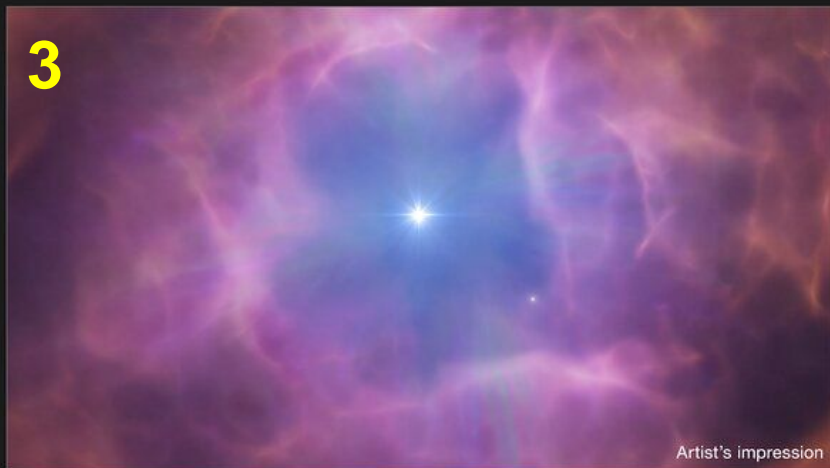
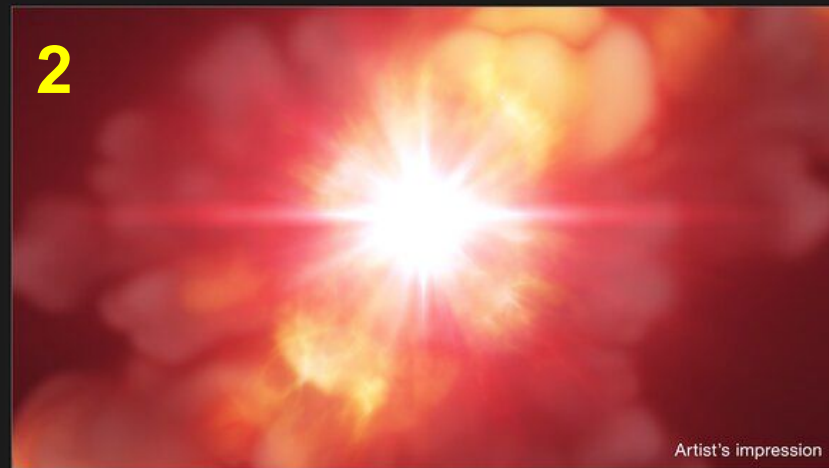
Merge with a binary companion!



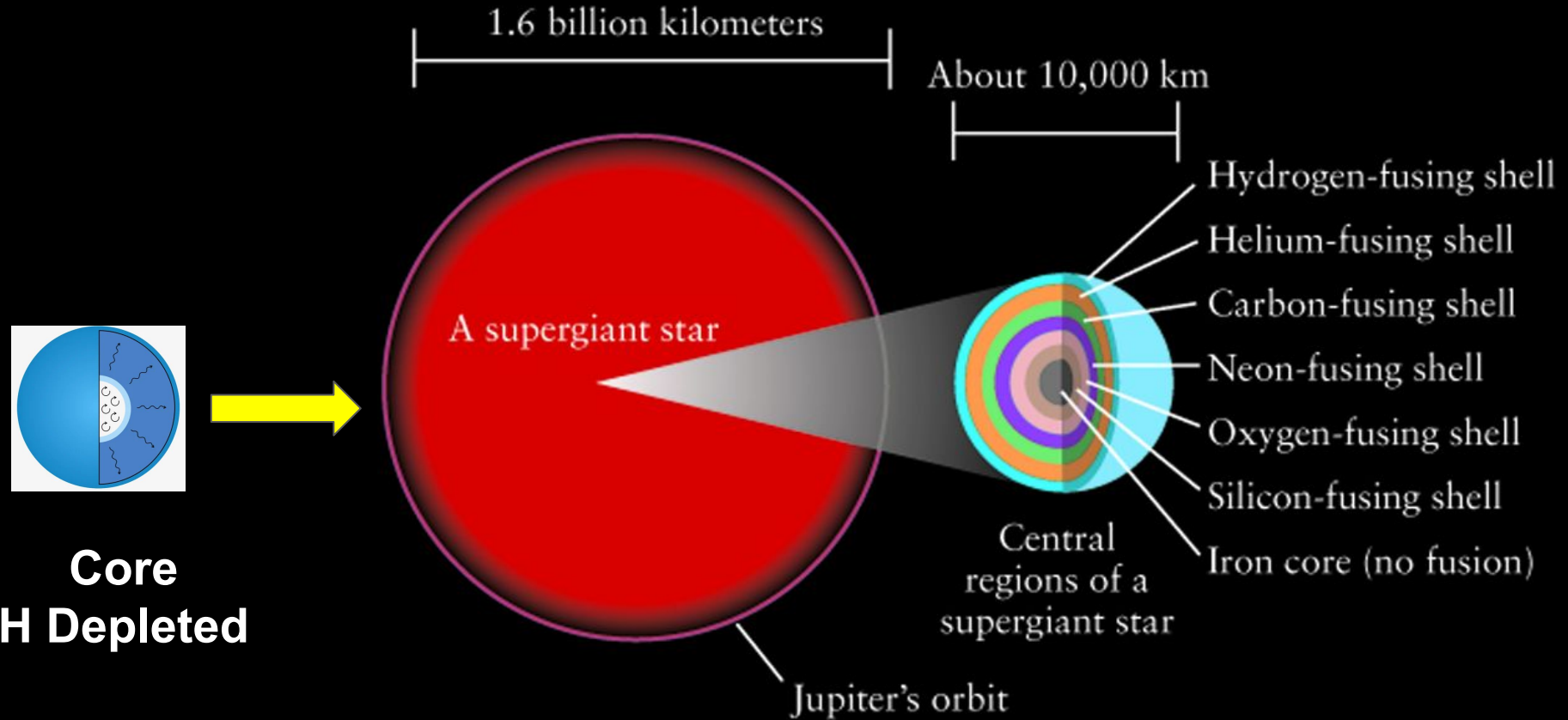
30 year Binary Orbit

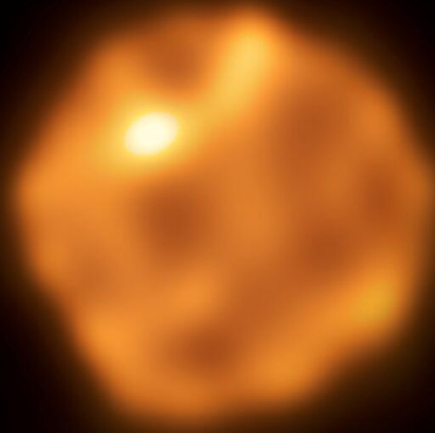


HD 148937 & Dragon's Egg Nebula

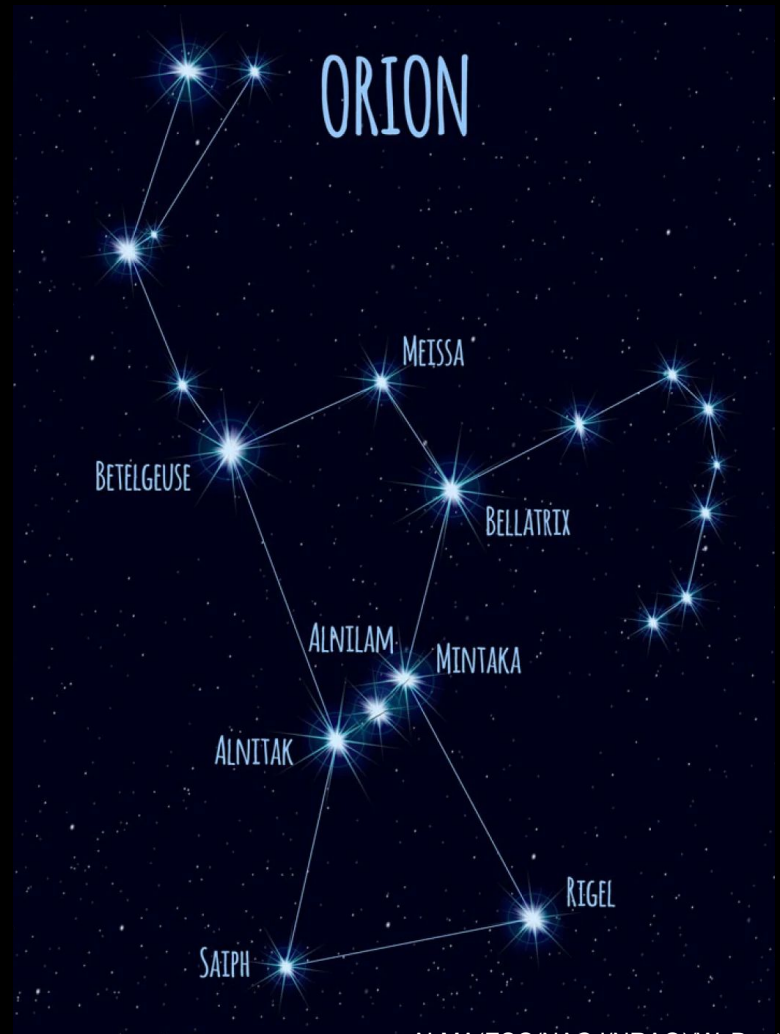


Late stage evolution



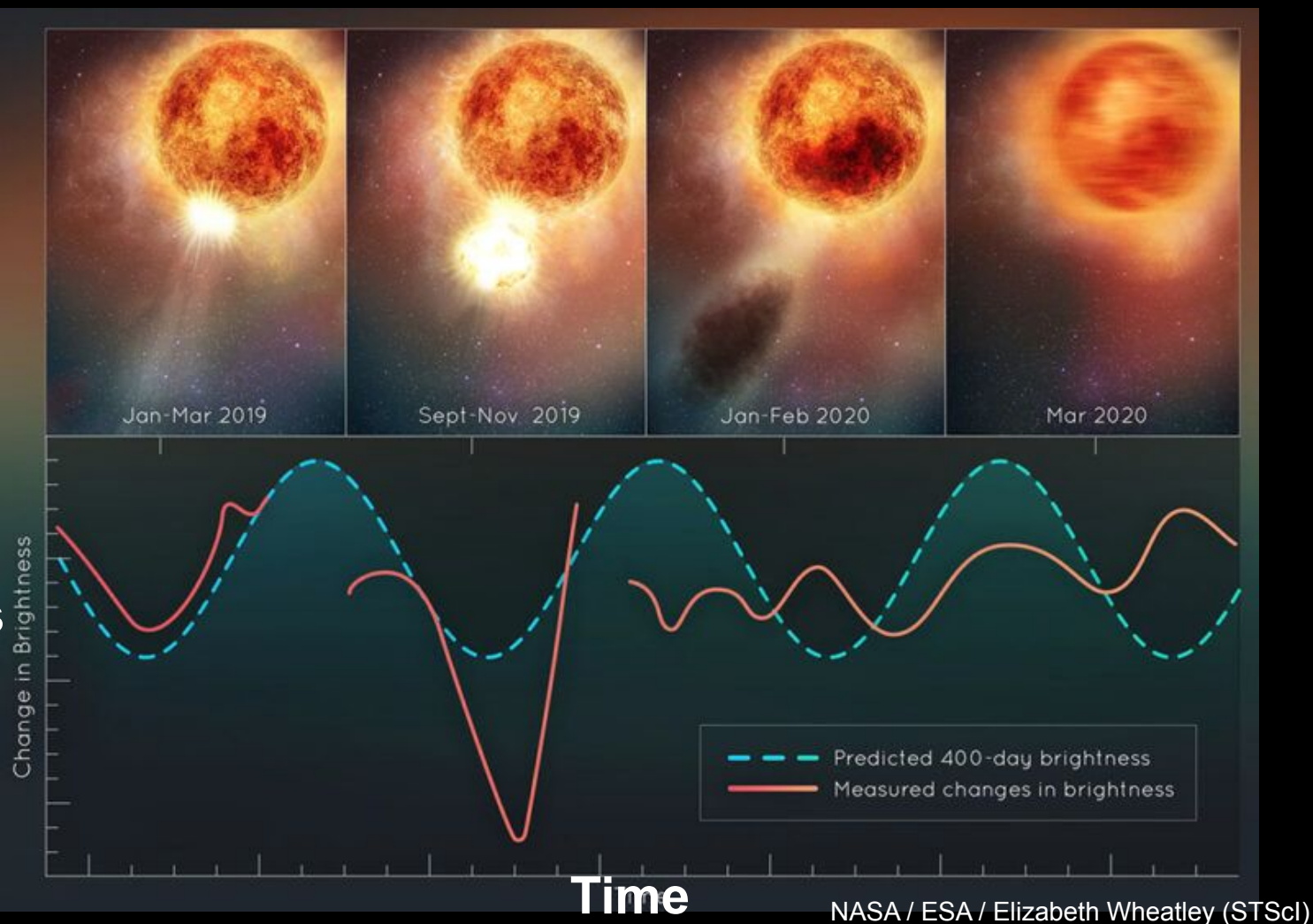


Betelgeuse



Great Dimming

Brightness

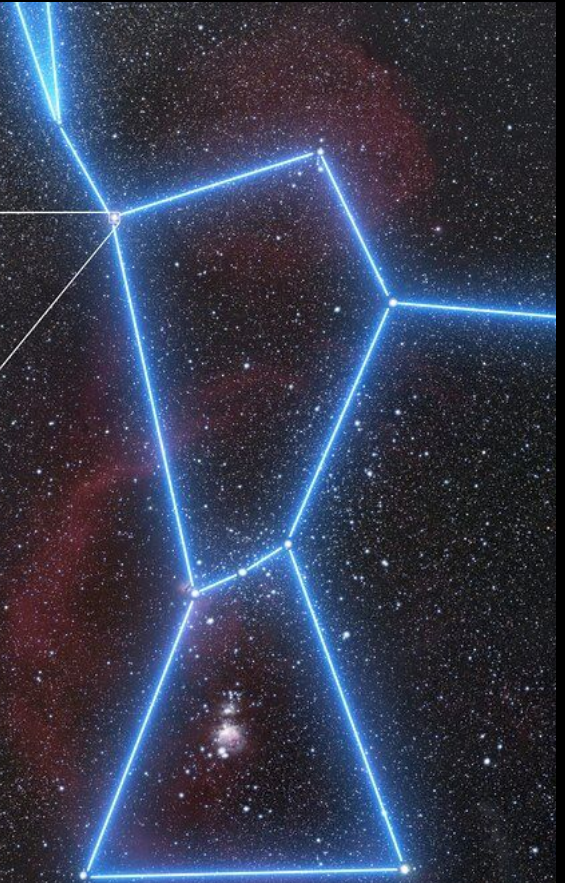
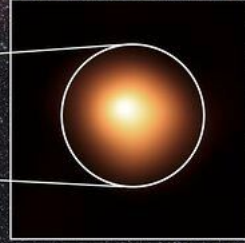
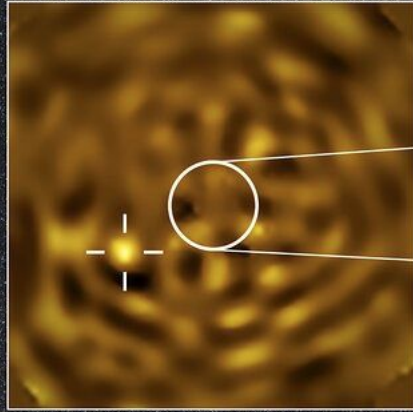


Betelgeuse Simulation

Betelgeuse is the 16th brightest star in the sky
Its surface is boiling violently



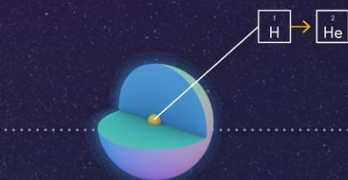
© MPA/Ha, Jing-Ze et al. 2024



Betelgeuse has a sister!

MASSIVE STARS: ENGINES OF CREATION

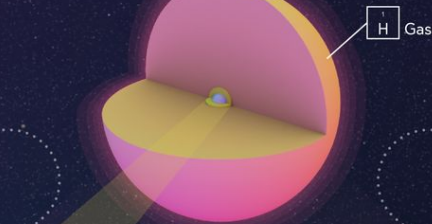
Through their cycle of formation, explosive "death" and reformation, massive stars (at least 8 times bigger than our sun) populate the universe with **new elements**, the raw material of all observable matter, including life.



1 MAIN SEQUENCE STAR

The star is fueled by hydrogen-to-helium fusion in its core.

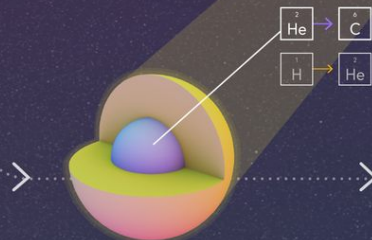
⌚ Stage lasts 7 million years



2 DEVELOPMENT OF MULTI-SHELLED CORE

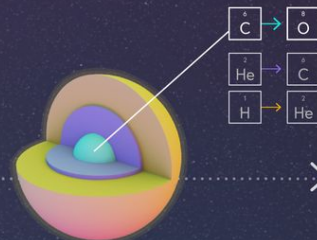
After hydrogen is exhausted in the core, other elements begin to fuse. Existing elements float above the denser center and form a multi-layered shell.

MULTI-SHELL CORE



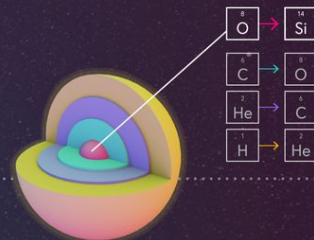
3 HELIUM-TO-CARBON FUSION

⌚ 500,000 years



4 CARBON-TO-OXYGEN FUSION

⌚ 600 years



5 OXYGEN-TO-SILICON FUSION

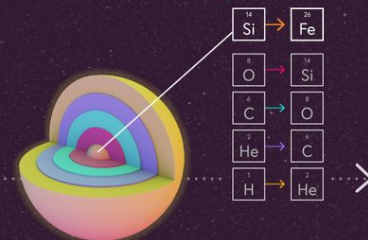
⌚ 6 months

2 He	6 C	7 N	8 O	9 F	10 Ne	11 Na	12 Mg
13 Al	14 Si	15 P	16 S	17 Cl	18 Ar	19 K	
20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn		
	26 Fe	27 Co	28 Ni	29 Cu	30 Zn		
		31 Ga	32 Ge	33 As	34 Se		
			35 Br	36 Kr	37 Rb		
				38 Sr	39 Y		
					40 Zr		

7 SUPERNOVA

The iron core cannot produce energy to balance the weight of the layered gas above. In less than one second, the core collapses and the upper layers rebound outward in a dramatic explosion, populating the universe with elements, the building blocks of creation.

⌚ 0.01 seconds



6 SILICON-TO-IRON FUSION

⌚ 1 day

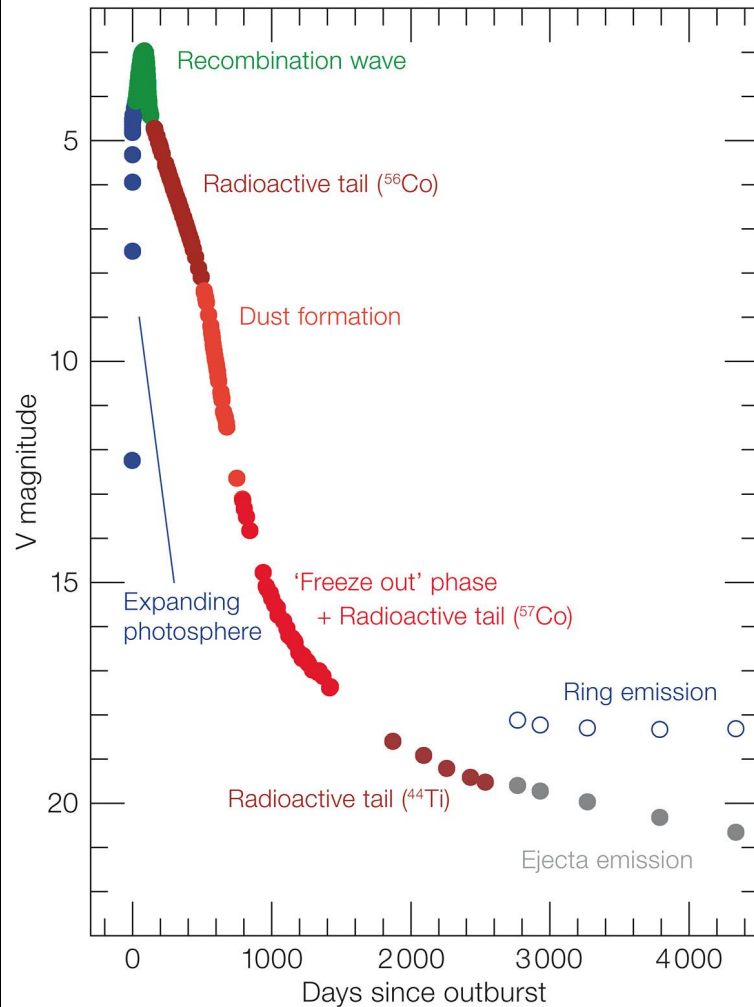
Image: NASA, ESA, Leah Hustak (STScI)

SN 1987A



Brightness

SN 1987A

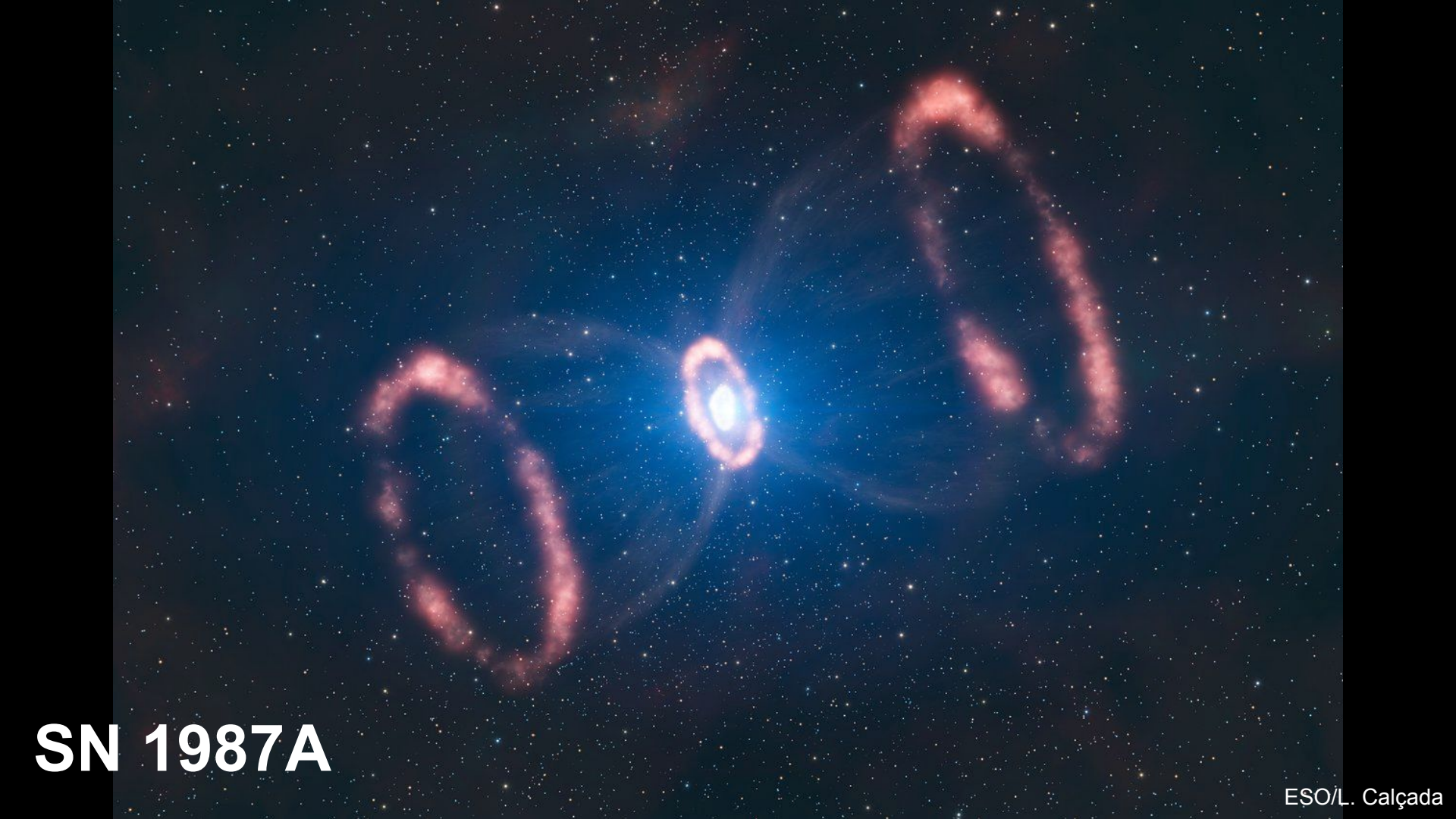


Time

SN 1987A

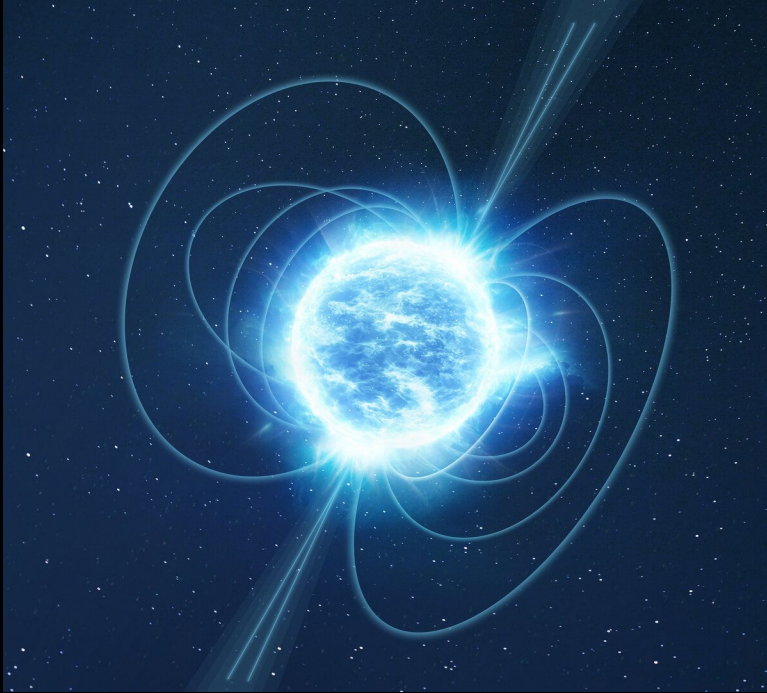
NASA, ESA, and R. Kirshner (Harvard-Smithsonian Center for Astrophysics)



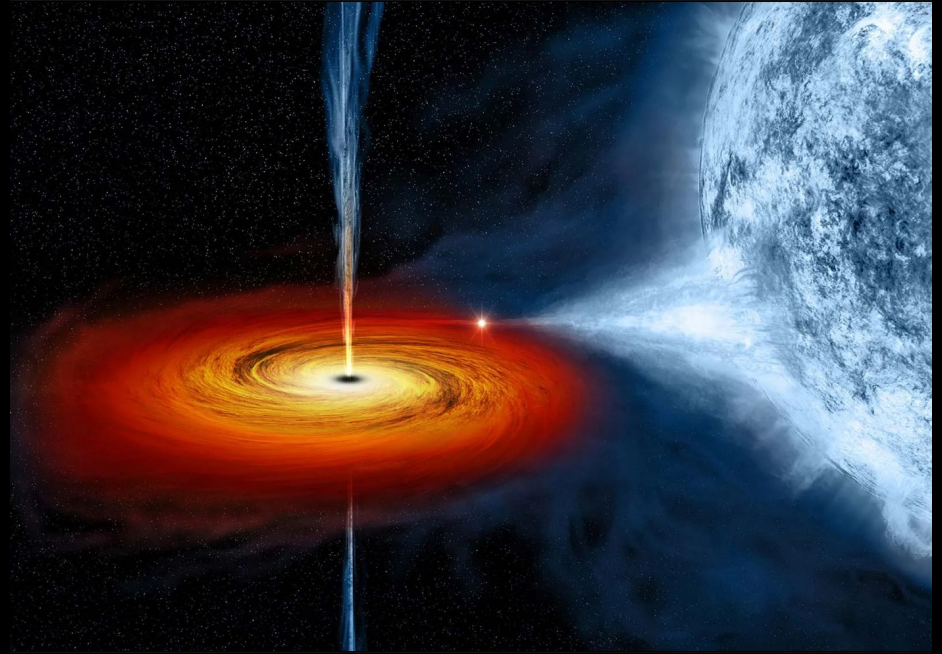


SN 1987A

Compact Remnants



Neutron Star



Black Hole

